Quiz

May. 11, 2026

多选题 2分

Three threads concurrently update a shared variable X, whose initial value is 0. Thread A performs X=X+2; thread B performs X=X-1; thread C performs X=0. Considering the possibility of race condition, what could be the possible outcome of X?

A 0

B 1

C 2

D -1

E -2

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多选题 2分

Which of the following is/are true about semaphore?

A. If the initial value of a semaphore is set to 3, this semaphore can be used by 3 processes to ensure only one process can enter the critical section.

B. Semaphores must be implemented using atomic instruction, otherwise concurrent calling sem_wait() and sem_post() may cause race condition.

C. To ensure bounded waiting, the wait list of a semaphore should be implemented as a stack.

D. Semaphore is an example of sleep-based locks for process synchronization.

E. When solving producer-consumer problem with semaphores, it is OK to swap the order of wait(&fill) and wait(&mutex).

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多选题 2分

Which of the following is/are true about safe state and deadlock state?

A If the system is in an unsafe state, it must also be in a deadlock state.

B If the system is in an unsafe state, there must be circular wait.

C If the system is in a safe state, it cannot be in a deadlock state.

D If the system is in a safe state, it may transit to a deadlock state in the future.

E If the system is in a deadlock state, it must also be in an unsafe state.

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多选题 2分

Which of the following statements about deadlock prevention and deadlock avoidance are true?

A Deadlock prevention aims to ensure that at least one of the four necessary conditions (mutual exclusion, hold and wait, no preemption, circular wait) cannot hold.

B Removing the "mutual exclusion" condition is always practical and commonly used in operating systems to prevent deadlocks.

C The Banker's Algorithm is a deadlock avoidance technique that requires processes to declare their maximum resource needs in advance.

D Deadlock avoidance guarantees that a system will never enter an unsafe state, but an unsafe state does not necessarily mean deadlock has occurred.

E Imposing a total ordering of resource types and requiring requests in increasing order prevents deadlock by removing the "hold and wait" condition.

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多选题 2分

Which of the following statements about the **resource-allocation graph** are correct?

A In a resource-allocation graph, a request edge is drawn from a thread to a resource type, and an assignment edge is drawn from a resource type to a thread.

B If the resource-allocation graph contains a cycle, then a deadlock must exist in the system, regardless of how many instances each resource type has.

C If each resource type has exactly one instance, then a cycle in the resource-allocation graph implies deadlock.

D The resource-allocation graph can be used to model which threads hold which resources and which resources threads are waiting for.

E If the resource-allocation graph has no cycles, the system cannot be in a deadlock state.

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主观题 3分

A system has **3 resource types**:

A with 10 instances

B with 5 instances

C with 7 instances

At time T_0 ,the system state is:

Process	Allocation (A B C)	Max (A B C)	Available (A B C)
P0	0 1 0	7 5 3	3 3 2
P1	2 0 0	3 2 2	
P2	3 0 2	9 0 2	
P3	2 1 1	2 2 2	
P4	0 0 2	4 3 3	

(a) Calculate the Need matrix.
(b) Show whether the system is in a safe state.
If yes, give at least one safe sequence; if not, explain why.
(c) Then, process P1 requests (1, 0, 2).
Check if the request can be granted immediately using the Banker's Algorithm.

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